Interface Test Plan

Interface: Launch and player selection

Interface Description: Console prompts select a human or autonomous computer player.

Test date: _____ Tester: _____ Build / commit: _____

Run java -cp /workspace/.setup/vjava-classes MyProgram in a UTF-8 terminal. Each case starts a new game unless stated otherwise. Expected results describe current behavior unless a usability target is explicitly identified.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Launch	UI-001	Compiled classes available.	Start MyProgram; inspect initial prompts.	No arguments	Displays how to play selection with human player and computer player options.	Not run
Human selection	UI-002	At player-choice prompt.	Enter 1; inspect board and next prompt.	1	Human starts at (1,1), 25 moves remain; enter how to move is shown.	Not run
Computer selection	UI-003	At player-choice prompt.	Enter 2; inspect difficulty menu.	2	Shows choose computer difficulty, 1. easy and 2. hard.	Not run
Menu clarity	UI-004	Fresh launch.	Read instructions without consulting source.	First screen	Usability target: numeric player choices and W/A/S/D instructions are discoverable. Current menu omits numeric labels and does not explain movement keys.	Not run
Other selection	UI-005	At player-choice prompt.	Enter another string.	human, blank, 3 (separate runs)	Current behavior: every choice other than exact 2 starts a human game.	Not run
Bad difficulty	UI-006	Computer selected.	Enter non-integer difficulty.	abc or blank	Current behavior: NumberFormatException terminates application; record as input-validation defect.	Not run
Other difficulty	UI-007	Computer selected.	Enter integer other than 1 or 2.	0 or 3	Current behavior: selects strategic behavior for any integer other than 1; no range validation.	Not run

Interface Test Plan

Interface: Board and movement

Interface Description: 20 x 20 emoji board, movement prompts and remaining-move feedback.

Test date: _____ Tester: _____ Build / commit: _____

Coordinates are zero-based. Reset the game for each independent case.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Board display	UI-008	Human game initialized.	Inspect board, player and outer walls.	1	20 rows, 20 tiles per row; player at (1,1), exit at (17,18); initial count 25.	Not run
Move right	UI-009	Human at (1,1).	Enter D.	D	Player moves to (1,2); old tile becomes floor; next display shows 24 moves.	Not run
Move directions	UI-010	Human at (1,1).	Enter D,S,A,W in order.	D, S, A, W	Positions become (1,2),(2,2),(2,1),(1,1); count becomes 21.	Not run
Letter case	UI-011	Human at (1,1).	Enter lowercase d then uppercase A.	d, A	Both accepted; player returns to start; count becomes 23.	Not run
Wall collision	UI-012	Human at (1,1).	Enter W then A.	W, A	Each prints cannot-move message; position and 25-move count unchanged.	Not run
Invalid input	UI-013	Human prompt active.	Enter unsupported or whitespace-padded input.	x, blank, dd, " d "	Prints u have to enter W, A, S, or D; no move is spent.	Not run
Input end	UI-014	Human prompt active.	Close standard input.	EOF	Current behavior: Scanner.nextLine throws NoSuchElementException; record missing graceful EOF handling.	Not run

Interface Test Plan

Interface: Items and doors

Interface Description: Tile effects, collection feedback and locked passage.

Test date: _____ Tester: _____ Build / commit: _____

Fixture cases use a separate test driver calling existing public methods; do not edit application source. Create Dungeon d, Player p(1,1), Timer t(25). For each fixture move p to the stated coordinate, call checkTile(p,t), then inspect output and state. These are planned tests, not executed results.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Red key	UI-015	Fresh fixture.	Place p on red key and call checkTile.	(9,13)	Red-key flag true; prints you picked up the RedKey.	Not run
Blue key	UI-016	Fresh fixture.	Place p on blue key and call checkTile.	(17,16)	Blue-key flag true; prints you picked up the BlueKey.	Not run
Locked doors	UI-017	No keys.	Call isValidMove for both doors; repeat after collecting matching keys.	Red (17,15); blue (17,17)	False without matching key; true with matching key. Wrong-color key alone does not unlock.	Not run
Door feedback	UI-018	Matching key collected.	Move onto matching door; call checkTile.	Red then blue in separate fixtures	Corresponding passed-door flag true; prints unlocked and went through message.	Not run
Health potion	UI-019	p.takeDamage(40) gives health 60.	Move onto potion; call checkTile.	(15,9)	Health becomes 85; health-potion message printed. Board tile is flask although item symbol is heart.	Not run
Speed potion	UI-020	Timer at 25.	Move onto speed tile; call checkTile.	(17,14)	Timer becomes 30; message says gained 5 extra moves. In game loop net gain is 4 after movement cost.	Not run
Trap	UI-021	Health 100.	Move onto trap; call checkTile.	(2,4)	Health becomes 80; trap message and health remaining: 80 printed.	Not run
Other tiles	UI-022	Fresh fixture.	Call checkTile on floor, monster and treasure separately.	(1,2), (2,8), (17,9)	Current behavior: no item message, health/key/timer change; monster and treasure have no implemented effect.	Not run

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Consumed tile	UI-023	Human enters trap from (1,4).	Move S then W then S.	Fixture position (1,4), health 100	First entry loses 20 health; after leaving, trap tile becomes floor, so return causes no further damage.	Not run

Interface Test Plan

Interface: Computer and game outcome

Interface Description: Autonomous turns, difficulty behavior and termination messages.

Test date: _____ Tester: _____ Build / commit: _____

Computer completion checks use bounded execution (for example timeout 10). Advanced outcome fixtures require a separate driver reproducing main-loop ordering with existing public APIs.

TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	PASS / FAIL
Easy computer	UI-024	Fresh launch.	Select computer, then easy; observe to game over.	2 then 1	Each turn announces Computer chose: W/S/A/D; no movement input needed; valid moves consume timer.	Not run
Hard computer	UI-025	Fresh launch.	Select computer, then hard; observe first turns and completion.	2 then 2	Deterministic greedy Manhattan strategy; from start first moves are S,S,S,D. Does not guarantee shortest path or escape.	Not run
Time exhausted	UI-026	Human starts at (1,1).	Alternate D,A twelve times, then D.	25 valid moves	At zero moves prints out-of-moves loss and terminates normally without another input request.	Not run
Health exhausted	UI-027	Fixture health 20; timer 10; adjacent trap.	Enter trap using main-loop operation order.	p at (1,4); S to (2,4)	Health becomes 0; health-loss message selected while time remains. Full console route is not asserted.	Not run
Escape	UI-028	Fixture has blue key, p at (17,17), timer 2.	Move D to exit using main-loop operation order.	Target (17,18)	Exit tile captured before overwrite; prints u escaped the dungeon. congrats.; loop ends.	Not run
Outcome priority	UI-029	Fixture can enter exit on final move, or trap on final move.	Exercise each final-move condition.	Exit with timer 1; trap with timer 1 and health 20	Exit takes priority over loss. When time and health both expire on non-exit tile, only time-loss message prints.	Not run